Mathematical Analysis — Midterm 2025, Row 1

Past exam · Răzvan

> **How to use.** The statement is on this page, exactly as in the official PDF (Class Materials → Past Exams → *midterm_2025_1.pdf*). Solve it on paper in **1h50min**, then turn to the solutions, which come after a red STOP page.

Midterm Test 2025 — Row 1

1. Let $A \subseteq \mathbb{R}$ be a bounded set. Prove that $\forall \varepsilon > 0,\ \exists x \in A$ such that $\sup(A) - \varepsilon < x$. 1p

2. Study the convergence of the following series: (a) $\displaystyle\sum_{n=1}^{\infty} \frac{a^n}{a+n},\ a > 0.$ (b) $\displaystyle\sum_{n=1}^{\infty} \frac{1 \cdot 3 \cdot \ldots \cdot (2n-1)}{2 \cdot 4 \cdot \ldots \cdot (2n)}.$ (c) $\displaystyle\sum_{n=1}^{\infty} \sqrt{n}\left(\sin\frac{1}{n} - \sin\frac{1}{n+1}\right).$ (d) $\displaystyle\sum_{n=2}^{\infty} \frac{1}{n\sqrt{\ln n}}.$ 4p

3. Compute $\displaystyle\int_0^{\infty} e^{-\sqrt{x}}\,dx.$ 1p

4. (a) Find the Taylor series around 0 for $\sin^2 x$. (1p)
   (b) Find the higher order derivatives $f^{(n)}(0)$, $n \in \mathbb{N}$, for the function (0.5p)

$$f(x) = \begin{cases} \dfrac{1-\cos x}{x^2}, & x \neq 0 \\ \dfrac{1}{2}, & x = 0. \end{cases}$$

5. Find the radius of convergence and the sum of the power series $\displaystyle\sum_{n=1}^{\infty} n^2 x^n.$ 1.5p

Time 1h50min. Marking starts from 1p. Total 10p.

⛔ STOP
SOLUTIONS BELOW

De aici încep rezolvările

Rezolvă mai întâi subiectul singur, cronometrat (1h50min), și abia apoi dă pagina. Toate paginile cu rezolvări au o bandă roșie în stânga.

SOLUTIONS — Midterm 2025, Row 1

1. Approximation property of the supremum

A is non-empty and bounded, so $s = \sup A$ exists (completeness axiom); s is the *least* upper bound of A.

Let $\varepsilon > 0$ and suppose, by contradiction, that no $x \in A$ satisfies $x > s - \varepsilon$, i.e. $x \le s - \varepsilon$ for every $x \in A$. Then $s - \varepsilon$ is an upper bound of A, and $s - \varepsilon < s$ — contradicting that s is the least upper bound.

$\Longrightarrow \exists x \in A$ **with** $\sup A - \varepsilon < x$ **(and of course** $x \le \sup A$**).** ■

2. Convergence of series

(a) Positive terms. Ratio test:

$$\frac{a_{n+1}}{a_n} = \frac{a^{n+1}}{a+n+1} \cdot \frac{a+n}{a^n} = a \cdot \frac{a+n}{a+n+1} \to a.$$

- $a < 1$: converges; $a > 1$: diverges (also $a_n \to \infty$).
- $a = 1$: the test is inconclusive; the series is $\sum \frac{1}{1+n}$, which diverges (harmonic).

$\Longrightarrow$ **converges** $\iff 0 < a < 1$**.**

(b) $a_n = \frac{(2n-1)!!}{(2n)!!}$, $\frac{a_{n+1}}{a_n} = \frac{2n+1}{2n+2} \to 1$ (inconclusive). Raabe–Duhamel:

$$n\left(\frac{a_n}{a_{n+1}} - 1\right) = \frac{n}{2n+1} \to \frac{1}{2} < 1.$$

$\Longrightarrow$ **diverges.**

Elementary alternative: $\frac{2k-1}{2k} \ge \frac{2k-2}{2k-1}$, which gives $a_n^2 \ge \frac{1}{4n}$, i.e. $a_n \ge \frac{1}{2\sqrt{n}}$, and $\sum \frac{1}{\sqrt{n}}$ diverges.

(c) By the Mean Value Theorem for $\sin$ on $\left[\frac{1}{n+1}, \frac{1}{n}\right]$:

$$\sin\frac{1}{n} - \sin\frac{1}{n+1} = \cos\xi_n \left(\frac{1}{n} - \frac{1}{n+1}\right) = \frac{\cos\xi_n}{n(n+1)}, \qquad \xi_n \in \left(\tfrac{1}{n+1}, \tfrac{1}{n}\right).$$

So the terms are positive and $a_n = \frac{\sqrt{n}\cos\xi_n}{n(n+1)}$. Limit comparison with $b_n = \frac{1}{n^{3/2}}$:

$$\frac{a_n}{b_n} = \frac{n^2\cos\xi_n}{n(n+1)} \to 1 \quad (\xi_n \to 0,\ \cos\xi_n \to 1).$$

$\sum n^{-3/2}$ converges ($p = \frac{3}{2} > 1$).

$\Longrightarrow$ **the series converges.**

(d) Positive, decreasing terms $\Longrightarrow$ Cauchy condensation:

$$2^k a_{2^k} = \frac{2^k}{2^k\sqrt{k\ln 2}} = \frac{1}{\sqrt{\ln 2}} \cdot \frac{1}{k^{1/2}} \quad \Longrightarrow \quad \text{diverges } (p = \tfrac{1}{2} \le 1).$$

$\Longrightarrow$ **diverges.**

General rule: $\sum \frac{1}{n(\ln n)^p}$ converges $\iff p > 1$.

3. Improper integral

Substitute $x = t^2$ ($t \ge 0$), $dx = 2t\,dt$; $x : 0 \to \infty \Longleftrightarrow t : 0 \to \infty$:

$$\int_0^\infty e^{-\sqrt{x}}\,dx = \int_0^\infty 2t\,e^{-t}\,dt.$$

By parts, $\int te^{-t}dt = -te^{-t} + \int e^{-t}dt = -(t+1)e^{-t}$, so

$$\int_0^\infty 2te^{-t}\,dt = 2\lim_{b\to\infty}\Big[-(t+1)e^{-t}\Big]_0^b = 2\big(0-(-1)\big) = 2,$$

using $(b+1)e^{-b} \to 0$.

$\Longrightarrow$ the integral converges and equals 2.

4. Taylor series and derivatives at 0

(a) Linearize first: $\sin^2 x = \frac{1-\cos 2x}{2}$, and $\cos y = \sum_{n\geq 0}(-1)^n \frac{y^{2n}}{(2n)!}$ for all $y \in \mathbb{R}$. With $y = 2x$:

$$\sin^2 x = \frac{1}{2} - \frac{1}{2}\sum_{n\geq 0}(-1)^n \frac{2^{2n}x^{2n}}{(2n)!} = \sum_{n=1}^{\infty}(-1)^{n+1}\frac{2^{2n-1}}{(2n)!}x^{2n} = x^2 - \frac{x^4}{3} + \frac{2x^6}{45} - \dots$$

$\Longrightarrow$ valid for all $x \in \mathbb{R}$ ($R = \infty$).

(b) For $x \neq 0$:

$$\frac{1-\cos x}{x^2} = \frac{1}{x^2}\sum_{n\geq 1}(-1)^{n+1}\frac{x^{2n}}{(2n)!} = \sum_{k\geq 0}(-1)^k\frac{x^{2k}}{(2k+2)!} = \frac{1}{2} - \frac{x^2}{24} + \dots$$

At $x = 0$ the series gives $\frac{1}{2!} = \frac{1}{2} = f(0)$. So f is the sum of a power series with $R = \infty$ on all of $\mathbb{R}$, hence $f \in C^\infty(\mathbb{R})$ and its coefficients are $\frac{f^{(m)}(0)}{m!}$. Comparing:

$$f^{(2k+1)}(0) = 0, \qquad f^{(2k)}(0) = (2k)!\cdot\frac{(-1)^k}{(2k+2)!} = \frac{(-1)^k}{(2k+1)(2k+2)}, \quad k \in \mathbb{N}.$$

$\Longrightarrow$ e.g. $f(0) = \frac{1}{2}$, $f''(0) = -\frac{1}{12}$, $f^{(4)}(0) = \frac{1}{30}$; all odd derivatives vanish (f is even).

5. Power series $\sum n^2x^n$

Radius: $\sqrt[n]{n^2} \to 1 \Longrightarrow R = 1$. At $x = \pm 1$, $|n^2x^n| = n^2 \not\to 0$, so the series diverges there. Convergence set: $(-1, 1)$.

Sum. Inside $(-1,1)$ we may differentiate term by term. Start from $\sum_{n\geq 0} x^n = \frac{1}{1-x}$:

$$\sum_{n\geq 1} nx^{n-1} = \frac{1}{(1-x)^2} \;\Rightarrow\; \sum_{n\geq 1} nx^n = \frac{x}{(1-x)^2}.$$

Differentiate again and multiply by x:

$$\sum_{n\geq 1} n^2x^{n-1} = \frac{d}{dx}\frac{x}{(1-x)^2} = \frac{(1-x)+2x}{(1-x)^3} = \frac{1+x}{(1-x)^3} \;\Rightarrow\; \sum_{n\geq 1} n^2x^n = \frac{x(1+x)}{(1-x)^3}.$$

$\Longrightarrow$ $R = 1$, and $\displaystyle\sum_{n=1}^{\infty} n^2x^n = \frac{x(1+x)}{(1-x)^3}$ for $|x| < 1$.

Results at a glance

Ex.	Answer	Tool
1	else $\sup A - \varepsilon$ would be a smaller upper bound	definition of sup

2	(a) iff $a < 1$; (b) diverges; (c) converges; (d) diverges	ratio; Raabe; MVT + comparison; condensation
3	2	$x = t^2$, by parts
4	$\sin^2 x = \sum (-1)^{n+1} \frac{2^{2n-1} x^{2n}}{(2n)!}$; $f^{(2k)}(0) = \frac{(-1)^k}{(2k+1)(2k+2)}$, odd $= 0$	linearization; coefficients $= \frac{f^{(m)}(0)}{m!}$
5	$R = 1$, $\frac{x(1+x)}{(1-x)^3}$	differentiate the geometric series twice